Sardar Patel Institute of Technology

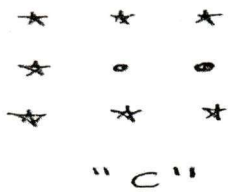
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

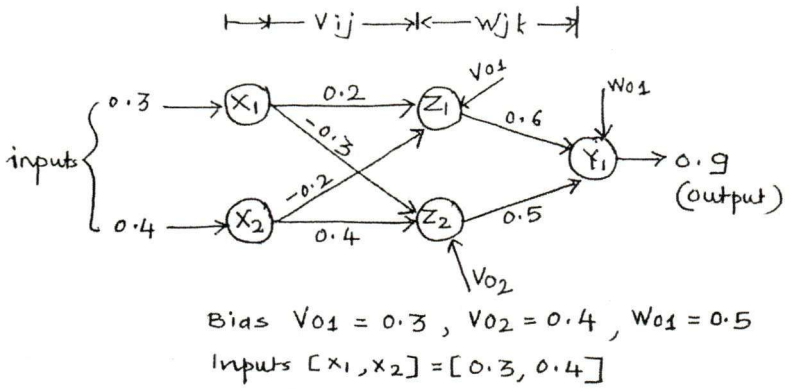
(Autonomous College Affiliated to University of Mumbai)

End Semester Examination
December 2025

Course Name: Artificial Intelligence & Soft Computing — Course Code: **CS303/CE303**
Class: T.Y.B.Tech. — Max. Marks: 100
Branch: COMP/CSE — Semester: V
Duration: 3 hours

Instruction:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary
(4) Mention the question number clearly while writing the answer

QNo	Question	Marks	CO-BL
Q1 a	Why does Rational thinking form the foundation of responsible and effective AI design?	10	CO1-3
Q1 b	What is the use of an auto-associative memory? Train an auto-associative network for input vector [−1 1 1 −1] and also test the network with same input vector. Test the auto-associative network with two missing and two mistake entries in test vector. Given : Activation Function $y_j = f(y_{inj}) = \begin{cases} 1 & \text{if } y_{inj} > 0 \\ -1 & \text{if } y_{inj} \le 0 \end{cases}$	10	CO3-3
Q2 a	Apply the Percepton Learning algorithm to classify the two-dimentional 3 x 3 input pattern as shown below. The pattern are C and D. The symbol " *" indicates data representation to be +1 and " " indicates data to be -1.  "C"  "D" For pattern C , the target is +1 and for D, the target is - 1. Assume initial weights as Zero [0] and learning rate as 1. Find the final weights.	10	CO2-3
Q2 b	Apply the backpropagation algorithm to find the final weights and bias for the following network. Learning Rate = 0.7 and target output = 0.9. Use Biploar Sigmoidal Activation Function.	10	CO2-3



Bias $V_{01} = 0.3$, $V_{02} = 0.4$, $W_{01} = 0.5$

Inputs $[x_1, x_2] = [0.3, 0.4]$

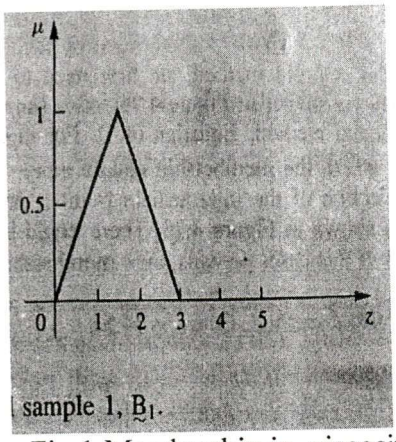
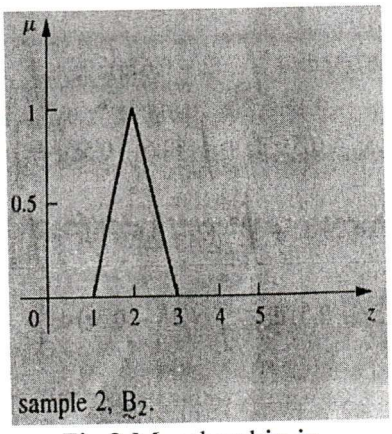
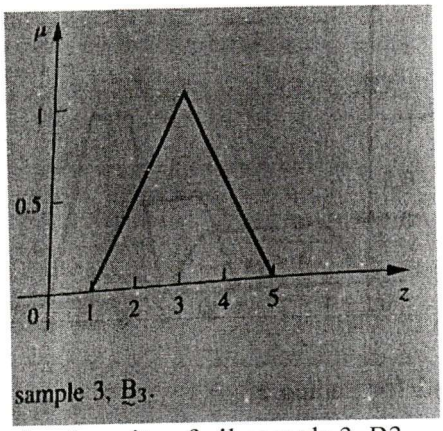
OR

Draw an architecture of Kohonen Self Organization Feature Map (KSOFM). Consider SOM with **4 cluster units** arranged linearly (units 1–4) and **5 input units**. Initial weights:

- w1=(0.9, 0.2, 0.1, 0.3, 0.4)
- w2=(0.7, 0.4, 0.3, 0.2, 0.6)
- w3=(0.4, 0.6, 0.5, 0.7, 0.2)
- w4=(0.1, 0.2, 0.8, 0.9, 0.3)

Input pattern x=(0.5, 0.5, 0.5, 0.5, 0.5)
Learning rate =0.5
Find the new weights for the winning unit

Q3 a	Explain with suitable examples Linearly and Non-linearly separable pattern classification	10	CO2-2
Q3 b	(i) Consider the fuzzy set G defined on the reference set U = {r1, r2 , r3, r4, r5} as given below: G = 0.475/r1 + 0.5/r3 + 1/r4 + 0.775/r5 Examine the five features for the given fuzzy set G. (ii) For a speed control of DC motor, the membership functions of series resistance, armature current and speed are given as follows: $R_{se} = \left\{\frac{0.4}{30} + \frac{0.6}{60} + \frac{1.0}{100} + \frac{0.1}{120}\right\}$ $I_a = \left\{\frac{0.2}{20} + \frac{0.3}{40} + \frac{0.6}{60} + \frac{0.8}{80} + \frac{1.0}{100} + \frac{0.2}{120}\right\}$ $N = \left\{\frac{0.35}{500} + \frac{0.67}{1000} + \frac{0.97}{1500} + \frac{0.25}{1800}\right\}$ Compute relation T for relating series resistance to motor speed. Perform max-min composition only.	10	CO4-3

Q4 a	Many products, such as tar, petroleum jelly, and petroleum, are extracted from crude oil. In a newly drilled oil well, three sets of oil samples are taken and tested for their viscosity. The results are given in the form of three sample fuzzy sets B1,B2, and B3, all defined on a universe of normalized viscosity, as shown in Figures 1-3.  Fig 1 Membership in viscosity of oil sample 1, B1  Fig 2 Membership in viscosity of oil sample 2, B2  Fig 3 Membership in viscosity of oil sample 3, B3 We want to find the most nearly representative viscosity value for all three oil samples, and hence find defuzzified value Z^* for the three fuzzy viscosity sets, using following methods 1. Center of sums, 2. Weighted average method 3. First of maxima 4. Last of Maxima	10	CO4-3
Q4 b	Design a fuzzy lighting controller system, in which the control system dim the bulb light automatically according to the environmental light. Assume that the inputs to the system are the environmental light x1 and the changing rate of the environmental light x2. While the output variable which represents the control value to the dimmer is DM. Consider the following assumption: Assume that x1 can be Dark (D), Medium (M), and Light (L) and its range between 120 and 220, with three membership functions: L(130,150), Π(130, 150 190,210), and Γ(190, 210) for D, M, and L respectively. x2 ranges between -10 and +10 and is divided into Negative-Small (NS), Zero (ZE), and Positive-Small (PS), with three membership functions:	10	CO4-4

	Λ (-20, -10, 0), Λ (-10, 0, 10), and Λ (0,10,20) for NS, ZE, and PS respectively. The output DM ranges between 0 and 10 and is divided into Very- small (VS), Small (S), Big(B), and Very-big (VB), with four membership functions: L(2,4), Λ(2,4,6), Λ(4,6,8), and Γ(6,8) for VS, S, B, and VB respectively. Evaluate the output for x1 = 125 and x2 = -6, having the following fuzzy rule base: | | D | M | L | | PS | B | S | VS | | ZE | B | B | S | | NS | VB | B | B |		
Q5 a	A fuzzy expert system for air-conditioning uses the rule: IF Room Temperature is Hot AND Humidity is High THEN Cooling Output = 0.5 × Temperature – 0.2 × Humidity + 5. (a) Identify whether this system uses Mamdani or Sugeno inference. (b) Explain the complete inference process for the identified FIS type. (c) Compare how the result would differ if the same system was implemented using the other type of FIS.	10	CO5-4
Q5 b	You are designing an ANFIS (Adaptive Neuro-Fuzzy Inference System) model to estimate the fuel efficiency of a car based on two input variables: engine load (L) and vehicle speed (V). Each input variable is represented using three linguistic terms: Low, Medium, and High. Input Variables: 1. Engine Load (L): Low, Medium, High 2. Vehicle Speed (V): Low, Medium, High Output Variable: Fuel Efficiency (F): The target output to be predicted. Draw the 5-layer ANFIS architecture corresponding to this system and explain the function of each layer, highlighting the roles of fuzzification, rule generation, normalization, and adaptive parameter tuning.	10	CO5-4